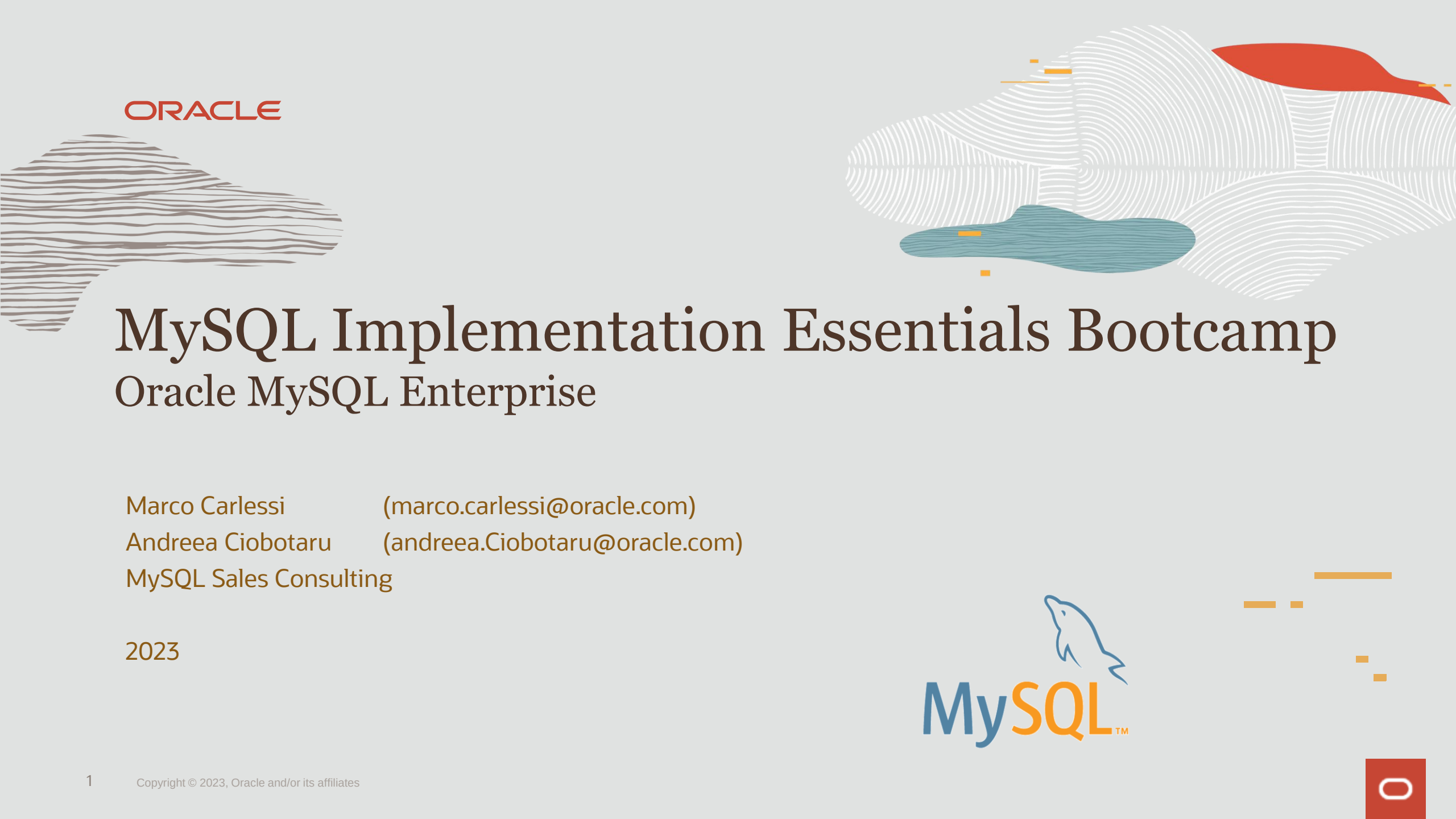




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MySQL Implementation Essentials Bootcamp

Oracle MySQL Enterprise

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MySQL Sales Consulting

2023



Program Agenda

- Introduction
- **Oracle MySQL Enterprise**
- Oracle Cloud Infrastructure overview
- Installation and Architecture
- Database Design
- Security
- Backup
- Replication
- High Availability
- MySQL Enterprise Monitor
- HeatWave

Licensing: MySQL is open source, but also...

All MySQL Binaries are available @ <http://www.mysql.com>

- Oracle MySQL is licensed under the GNU General Public License (GPL) and is free.
- MySQL Code is available :
 - ...but there are costs associated:
 - - Cost to integrate, deploy and secure
 - - Cost to certify, support and maintain
 - - Opportunity cost

GitHub

→ *Are you running MySQL efficiently?*

MySQL Commercial Products

Discussed in
this workshop

	MySQL Standard Edition	MySQL Enterprise Edition	MySQL Cluster CGE
Oracle Premiere Support	✓	✓	✓
MySQL Database Server	✓	✓	✓
MySQL Document Store	X	✓	✓
MySQL Connectors	✓	✓	✓
MySQL Replication	✓	✓	✓
MySQL Router	X	✓	✓
MySQL Partitioning	X	✓	✓
MySQL Workbench	✓	✓	✓
Storage Engine: MyISAM, InnoDB	✓	✓	✓
Storage Engine: NDB	X	X	✓
InnoDB Cluster	X	✓	✓
Oracle Enterprise Manager for MySQL	X	✓	✓
MySQL Enterprise Monitor	X	✓	✓
MySQL Enterprise Backup	X	✓	✓
MySQL Enterprise Security	X	✓	✓
MySQL Enterprise Scalability	X	✓	✓
MySQL Enterprise High-Availability	X	✓	✓
MySQL Cluster Manager	X	X	✓

Standard Edition
don't include

- Partitioning
- EE features
- NDB

Two MySQL «Flavors»

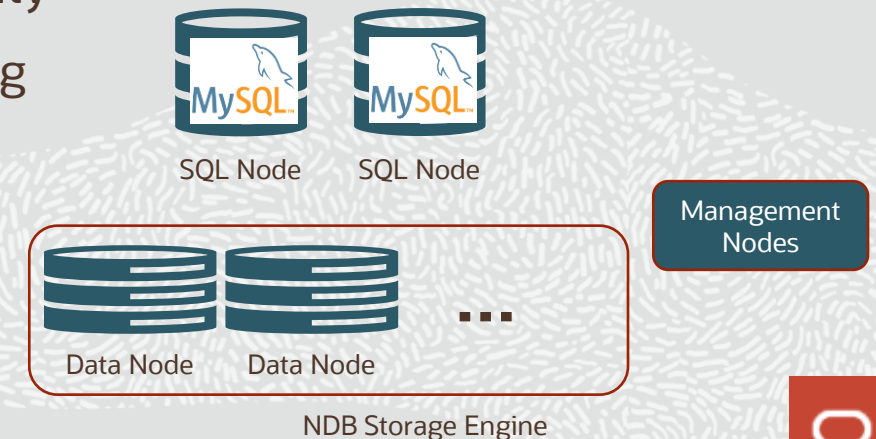
MySQL Enterprise Edition

- RDBMS
- Document Store capabilities
- Advanced security
- The most used by open source applications/projects
- Based on tablespaces



MySQL Cluster Carrier Grade (aka NDB Cluster)

- In-memory database
- RDBMS
- Key/Value NoSQL access
- High Availability (99,9999 SLA)
- High Scalability
- Auto Sharding



The Oracle MySQL business



Licensing



Support



Tools /
Advanced
Features



Cloud

MySQL Licensing rules

- ISV / OEM – Independent Software Vendor / Original Equipment Manufacturer
 - Embedded Database – Integrated Database and part of your product install
 - ISVs, OEMs, VARs and other distributors that:
 - Combine and distribute commercially licensed software with MySQL
- AND
- Do not wish to distribute the source code for the commercially licensed software under the GPL v. 2

⇒ Must enter into a commercial license agreement with Oracle

- Note: This rule applies if you embed the whole MySQL or just components (such as connectors)

<http://www.mysql.com/about/legal/licensing/oem/>

The Oracle MySQL business



Licensing



Support



Tools /
Advanced
Features



Cloud

Support: MySQL Enterprise Support

- Based on Metalink
 - <https://support.oracle.com>
- Largest MySQL engineering and support organization
- Backed by the MySQL developers
- World-class support, in 29 languages
- Hot fixes & maintenance releases
- 24x7x365
- Unlimited incidents
- Consultative support
- Global scale and reach



Get immediate help
for any MySQL issue,
plus expert advice

The Oracle MySQL business



Licensing



Support

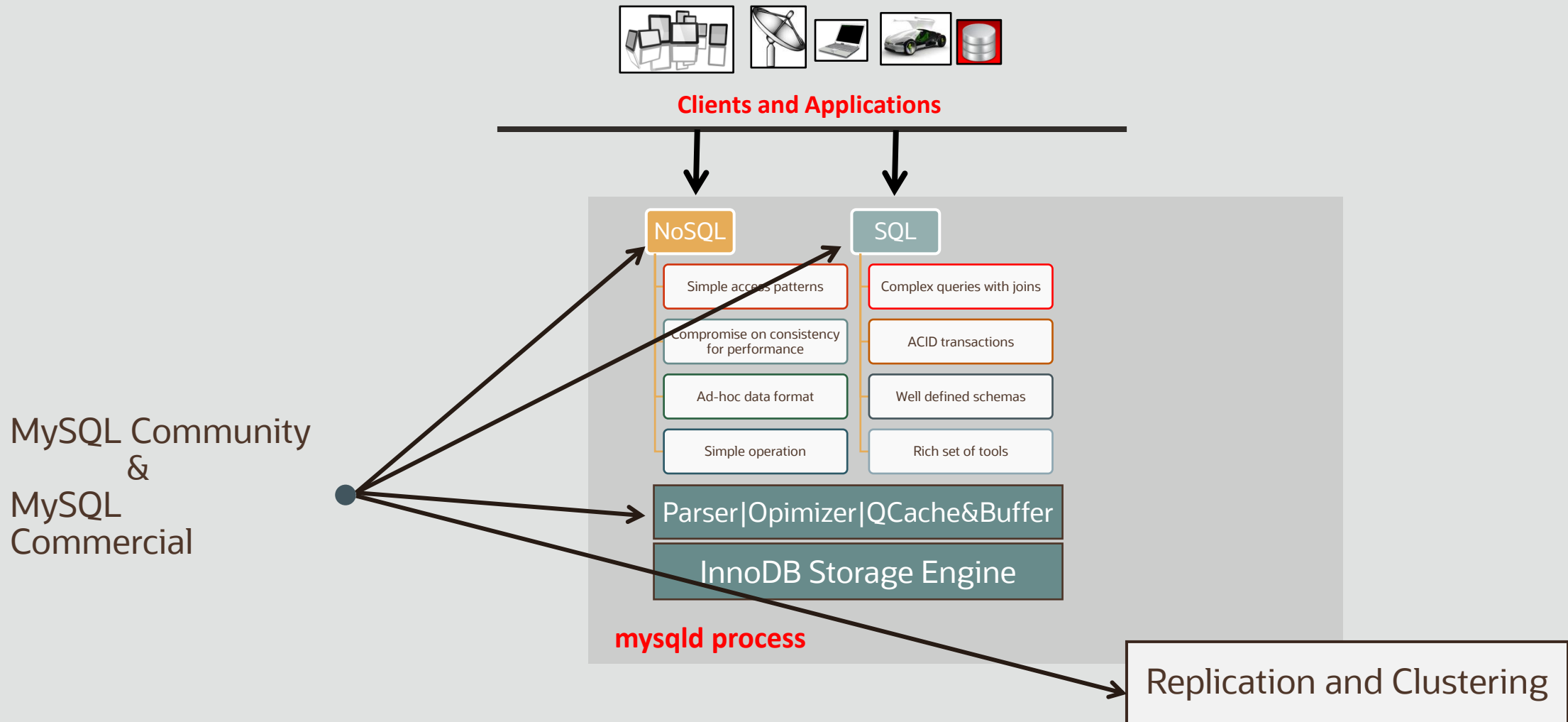


Tools /
Advanced
Features



Cloud

MySQL Community Edition



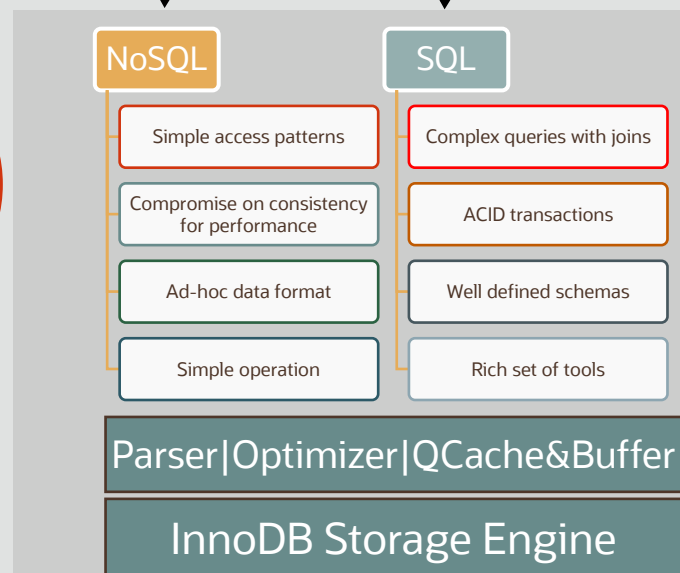
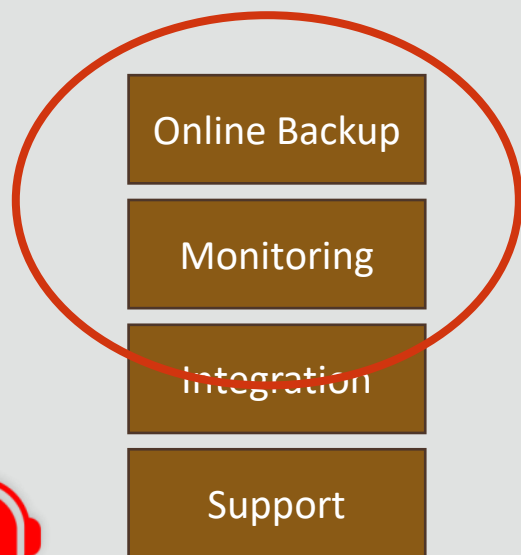
MySQL Enterprise Edition



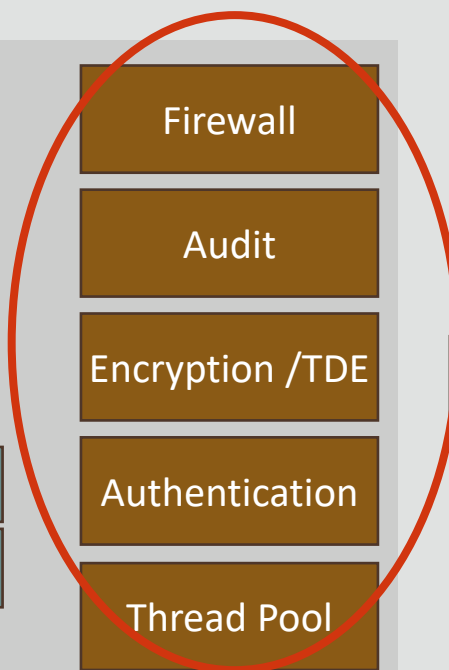
Clients and Applications

Tools

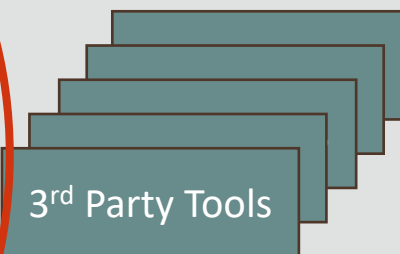
Plug Ins



mysql process



Replication and Clustering



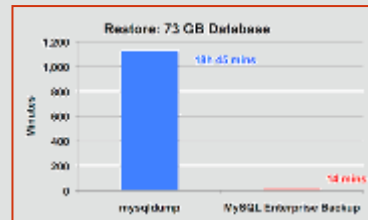
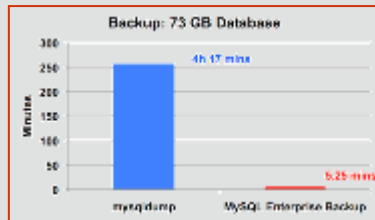
* All Plug-Ins and Tools are, 'free' as part of the Support Subscription



MySQL Enterprise Management Tools

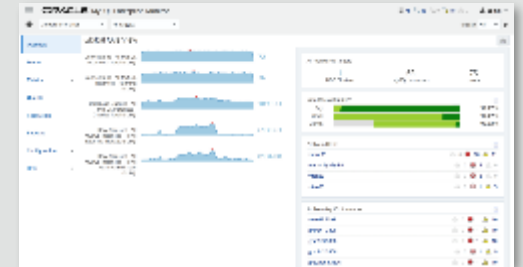
- MySQL Enterprise Backup

- Online, non-locking backup and recovery with best performances
- Advanced features (Incremental/Differential backups, compression, encryption...)
- Integration with MMS



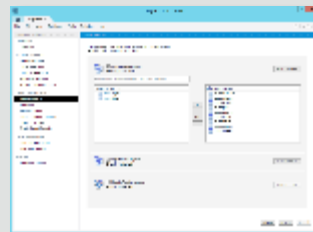
- MySQL Enterprise Monitor

- Real-time MySQL performance and availability monitoring
- Visually find & fix problem queries
- MySQL best practice advisors
- proactive and with forecasting



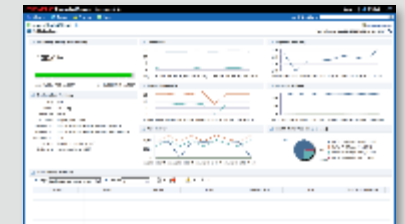
- MySQL Workbench Enterprise

- Database development, design and documentation
- Database migrations
- Database Administration



- Oracle Enterprise Manager Plugin for MySQL

- Availability and Performance monitoring
- Configuration monitoring
- All available metrics collected

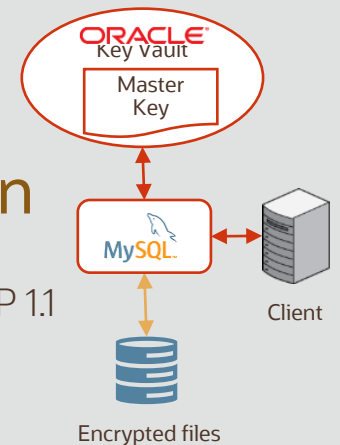


MySQL Enterprise Management Tools

- **MySQL Enterprise Thread Pool**
 - Improves scalability as concurrent connections grow
- **MySQL Enterprise Authentication**
 - Integrates MySQL with existing security infrastructures
 - Linux PAM, LDAP, Windows
 - Kerberos
 - FIDO (password less)
- **MySQL Enterprise Audit**
 - Out-of-the-box logging of connections, logins, and query (XML/JSON)
 - Simple to fine grain policies for filtering, and log rotation
- **MySQL Enterprise Firewall**
 - Out of policy database transactions detected and blocked



- **MySQL Enterprise Asymmetric Encryption**
 - MySQL encryption libraries
 - Key management, sign and verify data
- **MySQL Enterprise Transparent Data Encryption**
 - Data-at-rest encryption in the database
 - Master Key stored outside the database (KMIP 1.1 Compliant Key Vault)
- **MySQL Enterprise Masking**
 - Data Masking: String Masking, Dictionary Replacement
 - Random Data Generators: Range based, Payment Card, Email, SSN



ID	Last	First	SSN
1111	Smith	John	555-12-5555
1112	Templeton	Richard	444-12-4444

Masked View

ID	Last	First	SSN
2874	Smith	John	XXX-XX-5555
3281	Templeton	Richard	XXX-XX-4444

The Oracle MySQL business



Licensing



Support

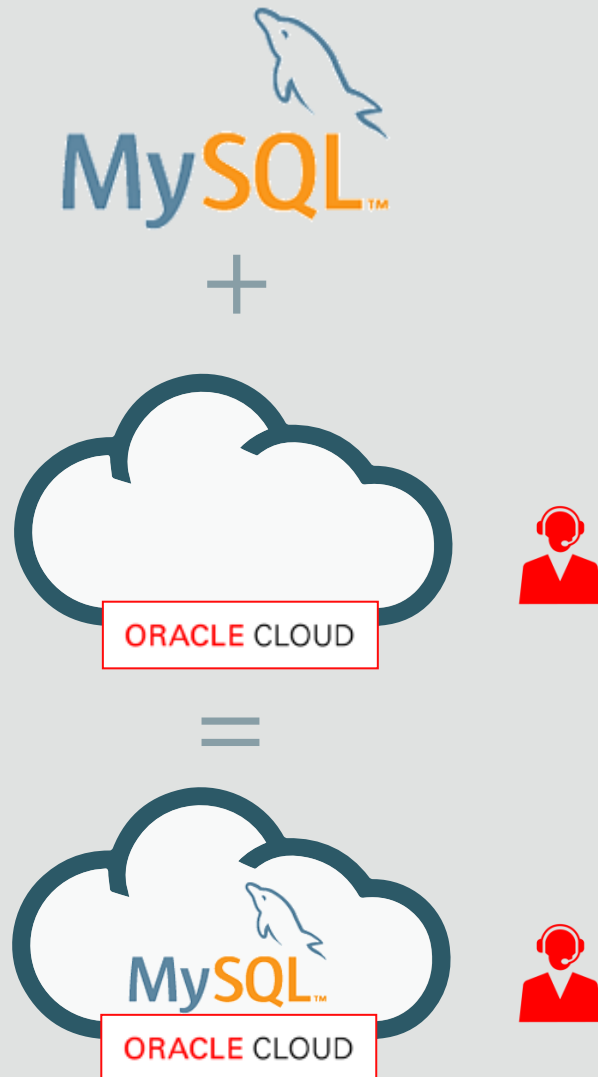


Tools /
Advanced
Features



Cloud

HeatWave: The Winning Combination



Simple,
Integrated and
Enterprise Ready

- HeatWave enables you to easily deploy modern applications powered by the World's Most Popular Open-Source Database
- Integrated with Oracle Cloud Infrastructure for quick development and deployment
- Based on Oracle's proven MySQL 8.0 Enterprise Edition

The Bottom Line

Performance

- Performance at scale
- Monitoring and tuning

TCO

- Reduced risk of downtime
- Improved Productivity

DevOps Agility

- Automated scaling & management
- Flexible, real-time backups

Peace of Mind

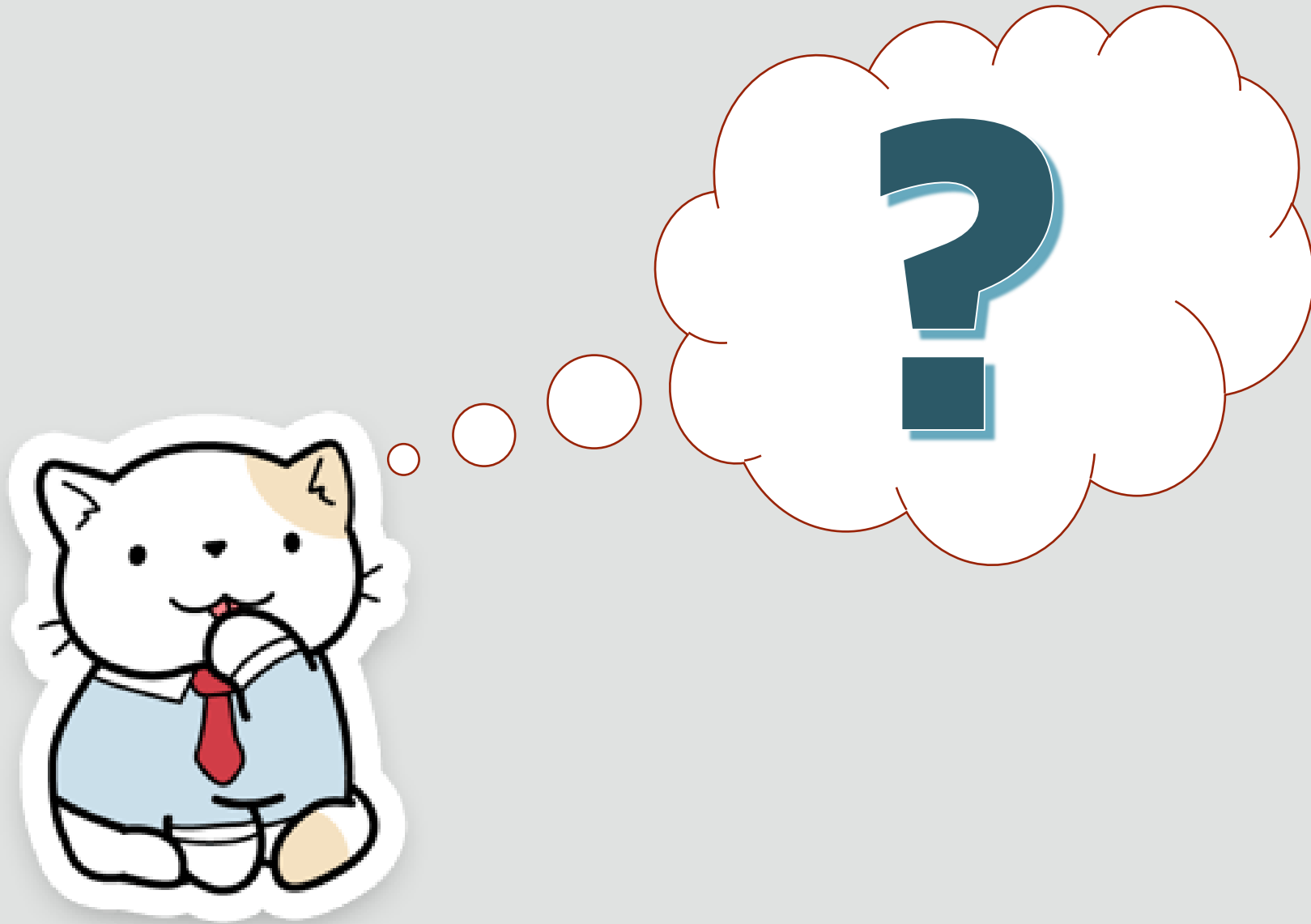
- Insurance
- Immediate help if/when needed

Risks

- Security & regulatory compliance
- Contact with MySQL team

Customer Satisfaction

- Application performance
- Application uptime





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